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ARTICLE

Do Regulatory Factors Explain China's Increasing Attractivity for Medical Device Development? A Comparison of the EU and Chinese Medical Device Landscapes

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Abstract

Medical device innovation is key to advancing healthcare and fostering economic development. As the global medical technology market expands, the capacity of regions and countries to attract and support innovation has become increasingly significant. This article examines the growing perception that China is becoming a more attractive environment for medical device development than Europe, focusing on key regulatory differences that may influence this shift. It compares the hierarchical structure of legislation, regulatory oversight bodies, and classification procedures, particularly for emerging technologies like artificial intelligence (AI). For instance, while Europe's decentralised system of notified bodies offers developers flexibility, China's centralised regulatory agencies and more adaptable classification system – alongside the absence of stringent AI-specific regulations like the EU's AI Act – may present a different set of trade-offs and facilitate faster market entry for innovative technologies. Despite these differences, the paper argues that regulation alone does not fully explain China's rising attractiveness. Other contributing factors, such as its vast market size, significant government support, and broader R&D policies, must also be considered. The analysis concludes that while regulatory structures are influential, they represent just one component of a multifaceted ecosystem that shapes the global geography of medical device innovation.

Keyword: medical device registration; MDR; AI-based medical devices; AI Act; China; EU

I. Introduction

Medical device innovation is important not only in a medical sense – promising novel methods for treating and preventing disease, but also in an economic sense. The medical device industry is worth billions of dollars and provides economic benefits in terms of employment and the fostering of a high value and innovate development ecosystem.[1] For these reasons the abilities of a country or region to attract and foster medical device innovation is considered important. In recent years, certain commentary has indicated

[1] MedTech Europe, *The European Medical Technology Industry in Figures 2023* (October 2023) <https://www.medtecheurope.org/wp-content/uploads/2023/10/the-european-medical-technology-industry-in-figures_2023.pdf> accessed 6 June 2025.

that the attractiveness of China as the place to undertake medical device innovation has been increasing relative to that of Europe.[2]

This paper looks at some salient differences between the Chinese and European medical device regulatory frameworks to discern how their respective designs is adding to this increased perception of China as a relatively attractive environment for medical device innovation. One critical distinction lies in, for example, the hierarchical position of the respective legislation giving rise to the regulatory framework in each jurisdiction. As will be discussed in Section II, in China such legislation enjoys a lower hierarchical position, arguably resulting in less certainty for medical device developers. Another major structural distinction, as analyzed in Section III, concerns the nature of the entities that are designed to provide regulatory oversight. In Europe, this function is conducted by decentralised "notified bodies," whereas in China, it is performed by more centralised governmental agencies. Given that the existence of multiple notified bodies has historically been perceived as an advantage in the European system because it provides extra capacity and choice, this centralised aspect of China's regulatory model may be considered less attractive to potential developers.

On the other hand, as Section V discusses, while both Chinese and the EU classification system applies a risk-based approach and shared similar classification criteria, the Chinese classification system seems to allow software and AI-based medical devices to obtain a lower, and thus less burdensome, classification more often. Furthermore, China does not yet have an equivalent to the AI Act of the EU which creates further significant regulatory burdens in Europe for all AI-based medical devices. China also foresees the possibility for an expedited approvals process for certain forms medical devices deemed to meet critical needs, such as those for rare diseases. This is a mechanism that is not as well-catered for within the EU's regulatory system.

In light of this mixed regulatory picture, the authors of this paper suggest (in Section VI) that one should not over-attribute the role of regulation in China's increasing attractivity as market for medical device innovation. Whist regulatory structures are undoubtedly important, they should be considered holistically alongside other factors that contribute to a market's relative appeal. These factors include, but are not limited to, market size, R&D policies, reimbursement frameworks, and the level of state support available.

II. The Position of Medical Device Regulation Within the Chinese Legal Hierarchy

In the EU, the regulatory framework governing medical devices is established by the Medical Device Regulation (MDR) 2017/745[3] and the In Vitro Diagnostic Device Regulation (IVDR) 2017/746[4]. Both regulations were introduced to address growing concerns over patient safety, device performance, and transparency, updating and replacing the Medical Device Directive (MDD) 93/42/EEC[5] and the In-vitro Diagnostic Directive (IVDD) 98/79/EC[6]

[2] Judy Zhao and Amanda Alkemark, 'China's Growing Demand for Healthcare Services and Medical Devices' (Business Sweden, 11 December 2024) <https://www.business-sweden.com/insights/articles/chinas-growing-demand-for-healthcare-services-and-medical-devices/> accessed 6 June 2025. See also: KPMG Huazhen LLP, *China Life Sciences Sector Overview and Outlook* (April 2025) <https://assets.kpmg.com/content/dam/kpmg/cn/pdf/en/2025/04/china-life-sciences-sector-overview-and-outlook.pdf> accessed 6 June 2025.

[3] Regulation (EU) 2017/745 of the European Parliament and of the Council of 5 April 2017 on medical devices, amending Directive 2001/83/EC, Regulation (EC) No 178/2002 and Regulation (EC) No 1223/2009 and repealing Council Directives 90/385/EEC and 93/42/EEC (MDR)

[4] Regulation (EU) 2017/746 of the European Parliament and of the Council of 5 April 2017 on in vitro diagnostic medical devices and repealing Directive 98/79/EC and Commission Decision 2010/227/EU (IVDR).

[5] Council Directive 93/42/EEC of 14 June 1993 concerning medical devices.

[6] Directive 98/79/EC of the European Parliament and of the Council of 27 October 1998 on in vitro diagnostic medical devices.

respectively, particularly in light of the divergent interpretations of the rules and several medical device incidents.[7] Within the EU's legal order, such legislative instruments represent secondary law, occupying a lower hierarchical rank than that of the EU treaties themselves. Such secondary legislation, however, is extremely authoritative, as it is binding upon the legal systems of all Member States. Consequently, national law, even those at the highest level, cannot contradict the requirements set out in such legislation.

In China, however, the situation concerning the legal framework relating to medical devices is somewhat different, at least in a formal sense. Regulation in this field has long been relying on a framework that centres around the Regulation on the Supervision and Administration of Medical Devices[8] (hereinafter referred to as China's MDR[9]). This is an "administrative regulation" adopted by the State Council, the highest governmental organ of the country, in 2000, with its most recent amendment in 2024. Within the Chinese legal system, "administrative regulations," whilst binding in nature, have a relatively lower legal hierarchical value than other sources of law, e.g., Constitution and national "laws" (equivalent to primary legislation at the national level in Europe and to "Legislative Acts" made by the EU itself),[10] which are enacted by the National People's Congress (NPC) and its Standing Committee.[11] As of now, no specific "law" has been passed in China exclusively addressing medical devices,[12] despite certain pertinent provisions linked to medical devices in other laws (such as the Standardisation Law,[13] the Advertising Law[14] and the E-commerce Law[15]). Thus, China's MDR, as an "administrative regulation," remains the most authoritative legal instrument specifically in regulating medical devices.

China's MDR lays down the general requirements and administrative procedures, covering the processes of development, manufacture, distribution and use of medical devices. It also establishes the oversight system and penalties throughout these phases. Based on this "administrative regulation," the implementation and enforcement are further detailed by a number of binding "departmental rules" issued by the competent

[7] European Commission, "New EU rules to ensure safety of medical devices" (5 April 2017) http://ec.europa.eu/commission/presscorner/detail/en/memo_17_848 accessed 14 April 2025.

[8] Regulation on the Supervision and Administration of Medical Devices (<医疗器械监督管理条例>, "China's MDR") (adopted on 4 January 2000, lastly revised on 6 December 2024), State Council of China <https://www.lawinfochina.com/display.aspx?id=44291&lib=law&SearchKeyword=&SearchCKeyword=>accessed 23 June 2025.

[9] In this paper, "China's MDR" is used as a concise abbreviation for the State Council's Regulation on the Supervision and Administration of Medical Devices, a legally binding instrument that ranks below primary legislation in China. It should be distinguished from "EU MDR" (Regulation 2017/745), which is directly applicable EU law binding all Member States.

[10] The EU Commission outlines what it sees as the EU's legal hierarchy on its website. It uses the term "legislative acts to denote regulations, directives and decisions adopted by an ordinary or special legislative procedure" (According to Art. 289 TFEU). For more see: https://eur-lex.europa.eu/EN/legal-content/glossary/eu-hierarchy-of-norms.html

[11] Legislation Law of the People's Republic of China (<中华人民共和国立法法>) (adopted on 15 March 2000, lastly amended on 13 March 2023), National People's Congress <http://en.moj.gov.cn/2023-12/15/c_948358.htm> accessed 23 June 2025. See Art. 99. See also: Xiaobo Dong and Yafang Zhang, *On Contemporary Chinese Legal System* (Springer Nature Singapore 2023) <https://link.springer.com/10.1007/978-981-99-2505-6> accessed 23 June 2025.

[12] It refers to the time of writing this paper.

[13] Standardisation Law of the People's Republic of China (<中华人民共和国标准法>) (promulgated on 29 December 1988, lastly amended on 4 November 2017), National People's Congress < https://lawinfochina.com/display.aspx?id=24182&lib=law > accessed 23 June 2025.

[14] Advertising Law of the People's Republic of China (<中华人民共和国广告法>) (promulgated on 27 October 1994, lastly amended on 29 April 2021), National People's Congress < https://www.lawinfochina.com/display.aspx?id=35521&lib=law > accessed 23 June 2025.

[15] E-Commerce Law of the People's Republic of China (<中华人民共和国电子商务法>) (promulgated on 31 August 2018, effective since 1 January 2019), National People's Congress < https://www.lawinfochina.com/display.aspx?lib=law&id=28792 > accessed 23 June 2025.

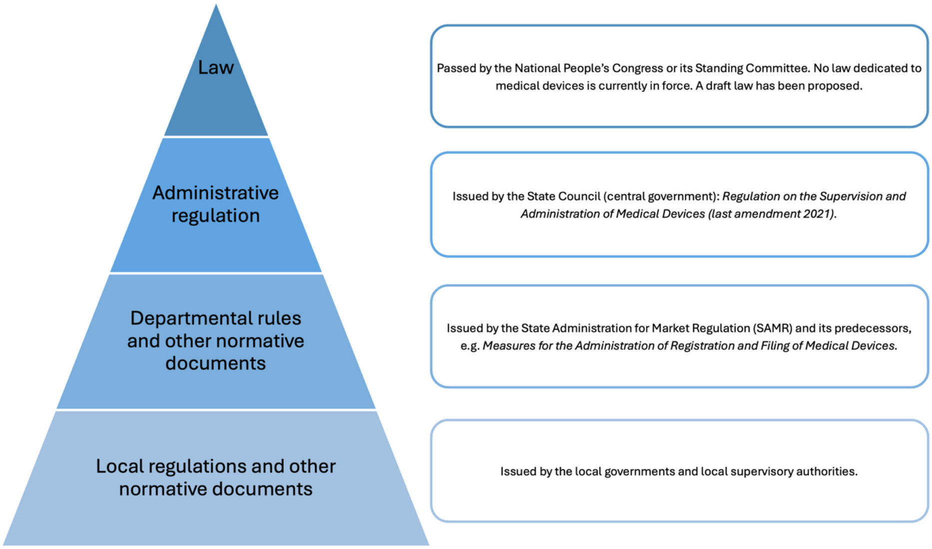

Figure 1. China's current hierarchical legal framework for medical devices.

authority for medical devices under the State Council, i.e., the State Administration for Market Regulation (SMAR), and its subordinate agency, the National Medical Products Administration (NMPA).[16] Such legally binding departmental rules include, for example, the Provisions for Medical Device Registration and Filing[17] and the Provisions for In-vitro Diagnostic Reagent Registration and Filing.[18] Additionally, the supervisory authorities at both central and local levels issue a wide array of regulatory materials, including binding rules and non-binding opinions, guidance and notices regarding various matters in relation to medical devices.[19] Complementing this hierarchical legal framework are national standards which, although not constituting a formal source of law in China, also play an important role in providing normative technical specifications in relation to the development, manufacture, distribution, use and oversight of medical devices. Taken together, these instruments contribute to a regulatory landscape in which no national "law" – in the formal legislative sense – exists specifically for medical devices. Instead, an "administrative regulation" issued by the central government serves as the backbone, supported by numerous fragmented normative documents of various legal status and effects (see Figure 1), along with technical standards.[20]

[16] The National Medical Products Administration (NMPA) of China was established in 2018 to replace its predecessor China Food and Drug Administration (CFDA).

[17] Provisions for Medical Device Registration and Filing (<医疗器械注册与备案管理办法>) (adopted on 26 August 2021), State Administration for Market Regulation < https://english.nmpa.gov.cn/2024-06/05/c_993242.htm> accessed 23 June 2025.

[18] Provisions for In-vitro Diagnostic Reagent Registration and Filing (<体外诊断试剂注册与备案管理办法>) (adopted on 26 August 2021), State Administration for Market Regulation < https://english.nmpa.gov.cn/2024-06/05/c_993311.htm> accessed 23 June 2025.

[19] This paper aims to focus on legal instruments issued by China's central government (including its departments) for regulating medical devices, as local laws and policies should not contradict with those at the central level. Also, the focus on central-level rules will avoid unnecessary complexity and excessiveness in the discussion.

[20] Jack Wong, "China: Medical Device Regulatory System," *Medical Regulatory Affairs* (3rd edn, Jenny Stanford Publishing 2022).

It is worth noting that this situation is expected to change in the upcoming years, with a national "law" being put on the agenda by the Chinese legislators. At the end of August 2024, the NMPA released a significant legislative proposal for medical devices, the draft Medical Device Management Law, for public consultation.[21] This draft law arguably aims to enhance the regulation of medical devices by consolidating fragmented rules into a single instrument with a higher legal status. Notably, the draft law emphasizes facilitating innovation and the development of the medical device industry as one of its stated goals.[22]

When comparing the hierarchical legal situation governing medical devices in China to that which exists in the EU, it is possible to make several observations. Within the EU, the regulatory framework benefits from a high degree of legal certainty, as it is established by Regulations – legal acts of the highest possible hierarchical position, subordinate only to the EU treaties themselves. This is not the case in China, where the existing legal instruments governing medical devices are not national "laws" passed by the NPC, but rather administrative regulations with a lower hierarchical status. Consequently, unlike in the EU, China's medical device regulations could conflict with other forms of legislation having a higher hierarchical position, which would then take precedence.

A significant result of this hierarchical arrangement is arguably the enhanced legal certainty afforded to medical device manufacturers under the EU law. This is because they have the knowledge that no new legislation will come into effect that will simply override the pre-existing legislation by virtue of having a higher hierarchical effect. The legal framework of medical devices (as it currently stands) will remain binding until it is explicitly abrogated by future legislation. There is additionally no risk that higher legislation may unintentionally override elements of the EU MDR because of an incidental overlap in subject matter. This stands in contrast to the situation in China where stakeholders involved in the medical device production chain must be aware of the potential that other national "laws" could, at least in theory, conflict with and take precedence over the existing administrative regulations. As is discussed above, however, this specific vulnerability may be resolved in the near future with should a binding national "law" on medical devices be enacted.

III. The Differences in Regulatory Actors in China and the EU

Whilst it is a common complaint in European commentary that the regulatory structure surrounding the MDR is too complex,[23] the organisational structure of the entities responsible for device approval is, in reality, relatively simple in Europe—particularly when compared to that in China. In the EU, despite the fact that the authorization process for medical devices is not centralized, it is clearly established that 'notified bodies' are responsible for granting regulatory approval to medical devices.[24] These notified bodies are typically private entities (though public ones exist too) that have been officially

[21] The General Department of the NMPA, 'Publicly solicits opinions on the "Medical Device Management Law of the People's Republic of China (Draft for Comment)' (<国家药监局综合司公开征求《中华人民共和国医疗器械管理法（草案征求意见稿)》意见>)" (28 August 2024) < https://www.nmpa.gov.cn/xxgk/zhqyj/zhqyjylqx/20240828155857156.html?type=pc&m=> accessed 23 June 2025.

[22] The draft Medical Device Management Law 2024, Art. 1 and Art. 4–6.

[23] Arthur Arnould, Rita Hendricusdottir and Jeroen Bergmann, "The Complexity of Medical Device Regulations Has Increased, as Assessed through Data-Driven Techniques" (2021) 3(4) *Prosthesis* 314 <https://doi.org/10.3390/prosthesis3040029> accessed 23 June 2025. See also: Liga Svempe, "Exploring Impediments Imposed by the Medical Device Regulation EU 2017/745 on Software as a Medical Device" (2024) 12(1) *JMIR Medical Informatics* e58080 <https://www.jmir.org/2024/1/e58080> accessed 23 June 2025; Yu Han and others, "More than Red Tape: Exploring Complexity in Medical Device Regulatory Affairs" (2024) 11 *Frontiers in Medicine* 1415319 <https://doi.org/10.3389/fmed.2024.1415319> accessed 23 June 2025.

[24] Regulation (EU) 2017/745, Article 52 and 53

designated by a national authority and subsequently notified to the European Commission to perform conformity assessments under the EU medical device framework.[25] This system provides manufacturers in the EU with a choice as to which notified body will be used for the conformity assessment.[26] Whilst the EU system allows notified bodies to pursue certain specialisms and structure of their approval process differently,[27] medical device manufacturers in Europe benefit from a high degree of legal certainty that once a medical device receives approval from a notified body anywhere in the EU, it can be lawfully marketed through the EU' single market.

By contrast, in China, the authority to approve medical devices is vested exclusively in the government, specifically the competent central and local regulatory agencies. This system is defined by a dual structure of strict hierarchical control and significant functional division, underscoring the centralised nature of regulatory oversight to ensure, in theory, that all regulatory actions are subject to uniform governmental oversight. A comparison with the EU reveals several distinct features of such an institutional arrangement. First, both the central NMPA and its local agencies are granted regulatory power to involve in the approval process for medical devices, but they have different roles and areas of responsibility – partly to reduce the workload of the central medical device regulator.[28] For example, the central NMPA is only responsible for reviewing and approving the domestic medical devices classified as Class III and all the imported medical devices, while Class II and Class I medical devices are respectively certified by province-level and city-level agencies of the NMPA.[29] The applicant must approach the appropriate regulatory authority as determined by the classification of medical device and the geographical location of relevant business activities. Despite this division, it is obvious that the power to approve medical devices remains exclusively in the hands of government agencies (unlike Europe where privatised "notified bodies" are permitted to make decisions).

Second, the organisation of regulatory bodies across central and multiple local levels is hierarchical, with the NMPA at the highest level, exercising authority over subordinate agencies nationwide.[30] The hierarchical relationship is replicated at multiple regional levels, where lower-tier bodies are subject to supervision by both their immediate superiors and the central authority.[31] While the decision-making competences for Class II and Class I medical devices are delegated to bodies that act at a more local level, superior regulatory bodies retain the power to inspect, conduct inquiries and mandate corrective actions concerning the decisions and activities of their subordinates.[32] Consequently, medical device manufacturers must engage with the legally mandated authority for placing their devices in the Chinese market, rather than having the freedom to choose whom to communicate with.

[25] Regulation (EU) 2017/745, Article 35 and 42

[26] European Commission, 'Notified bodies' (Single Market Economy) <https://single-market-economy.ec.europa.eu/single-market/goods/building-blocks/notified-bodies_en> accessed 19 June 2025.

[27] The EU's NANDO (New Approach Notified and Designated Organisations) database is the official public record of all notified bodies (https://webgate.ec.europa.eu/single-market-compliance-space/notified-bodies). For each notified body listed, the database provides the precise scope of notified items using a series of codes that correspond to different categories of medical devices. This is direct and practical evidence that each notified body has a legally defined and limited area of expertise.

[28] Wenbo Liu and others, 'Review and Approval of Medical Devices in China: Changes and Reform' (2018) 106(6) *Journal of Biomedical Materials Research Part B: Applied Biomaterials* 2093.

[29] China's MDR 2024, Art. 15 and 16. See also Wong (n 20).

[30] China's MDR 2024, Art. 3

[31] China's MDR 2024, Art. 4

[32] China's MDR 2024, Art. 74

Such an institutional arrangement in China contrasts sharply with that of the EU, where notified bodies can specialise and tailor their approval procedures to varying types of medical devices. In this environment, entities involved in the production of medical devices are able to "shop around," choosing the type of notified body that best aligns with their specific needs. In China, however, this discretion is absent; the designated regulatory authority is determined by the classification of device and where processes in its production or distribution are to be deployed. To further clarity this comparison, the following section will elaborate on the differences in responsibilities among these regulatory agencies in EU and China's classification-based system of certifying medical devices.

IV. Beyond Certification: China's Multi-Tiered Regulatory Approvals for Medical Devices

In terms of regulatory procedures, the EU and China share important structural similarities, most notably an approach that is based on the classification of different types of medical devices in terms of the varying perceived risks. That is to say, different procedures are determined by the different risk levels associated with the medical devices, determined by similar key criteria including duration of use, degree of invasiveness and energy sources.[33] For instance, in both jurisdictions, low-risk (Class I) devices are subject to lighter regulatory requirements. In the EU, these devices generally receive a CE marking through a self-certification process without the involvement of notified bodies, provided that technical documentation is maintained and available for inspection.[34] Similarly, in China, Class I devices are not required to undergo pre-market approval with the NMPA but only need to be filed with provincial or municipal regulatory authorities, with manufacturers responsible for ensuring conformity and maintaining technical documentation for inspection.[35] Furthermore, both systems incorporate clinical evaluation as an essential component of the conformity assessment process. Specifically, both frameworks allow certain low-risk devices to be exempted from mandatory clinical evaluation if sufficient scientific evidence is available,[36] thereby reducing regulatory burden and expediting market access. It mirrors the EU's mandate for clinical investigations of Class III and implantable devices, which must be approved by Member States, while the terminological and procedural details differ.[37] For instance, the term "clinical trial" in the EU law is specifically used in the context of medicine products, differing from China.[38]

One prominent divergence worth noting is that China's medical device regulatory framework offers a few special approval procedures based on a device's level of innovation,

[33] In China, medical devices are classified into three risk-based categories: Class I (low risk, general controls), Class II (moderate risk. stricter controls), and Class III (high risk, strictest oversight). Similarly, the EU classifies medical devices under Class I (low risk), IIa (medium-low), IIb(medium-high), and III (high). While the specific classification criteria and rules differ between the two regions, a comparative study on the general classification of medical device falls outside the scope of this paper. Instead, this paper focus on the distinctions in regulatory procedures in Section IV and classification of AI-based medical devices in Section V. Detailed Chinese Medical Device Classification is provided in Annex I.

[34] Regulation (EU) 2017/745, Art. 52(7).

[35] China's MDR (2024), Art. 15.

[36] China's MDR 2024, Art. 25 and 27; Regulation (EU) 2017/745, Art. 61(4).

[37] Regulation (EU) 2017/745, Art. 61(4).

[38] In the EU law, the term "clinical trial" is defined in the Clinical Trials Regulation (EU) No 536/2014 which exclusively concerns medicinal products. In the context of medical device regulation, the EU's MDR uses the term of "clinical investigation" when referring to studies for assessing the safety or performance of a device. By contract, Chinese existing regulatory framework officially and commonly uses the term "clinical trial" for equivalent studies on medical devices.

urgency and scarcity, which could be accelerated. These special approval pathways include conditional approval for rare-disease or emergency devices with ongoing study requirements; an innovative-products route with early engagement and priority review; a priority-review track for urgently needed clinical technologies; and an emergency-approval option (by NMPA only) when no equivalent device exists.[39] The accelerated pathways are as important as the routine one given that medical device regulators have the mission not only to protect patients from unsecure medical products and promote public health for the whole society,[40] but also to play a crucial role in influencing the innovation and competitiveness in this sector.[41] It has been reported that many industry stakeholders increasingly leverage China's special approval pathways to accelerate medical-device market entry within the regulator's efficiency-focused framework.[42] In contrast, however, this is not the case within the EU's medical device regulatory framework. Scholars have expressed a concern that the EU MDR risks delaying patient availability of truly innovative and high-risk devices compared with ad-hoc accelerated approval pathways in other jurisdictions,[43] such as in China. Likewise, a similar critic exists in the European medical device industry, calling for creating dedicated and accelerated assessment pathways for innovative and emerging medical technologies within MDR and IVDR frameworks.[44]

At the post-certification stage, however, the divergence between the EU and Chinese regulatory systems becomes most pronounced. In Europe, the CE marking represents in single point of an approval; once obtained, a medical device can be placed on the market anywhere in the EU without requiring separate licences for its manufacture, storage or distribution. China's MDR, by contrast, adopts a lifecycle approach that stipulates distinct requirements and additional forms of regulatory approval for activities following initial device certification. It mandates separate licences for manufacturing, distribution and certain associated business activities (such as online retail), and the certain use of medical devices (such as in hospitals) – phases for which no specific authorisation is needed in the EU. The requirements and approval procedures for each of these phrases also vary, depending on the risk classification of the medical device. The following provides an overview of the additional forms of approval processes required in China after the relevant certification has been obtained.

In contrast to the EU, where manufacturing is generally unrestricted post-authorisation, China's regulatory regime imposes additional formalities. In China, manufacturing a registered Class II or III medical device requires a separate permit beyond general market authorisation. Applications must be submitted to the provincial-level supervisory authority with jurisdiction over the manufacturer's location.[45] Class I

[39] Medical Device Registration Provisions 2021, Arts. 61,62, 68, 73 and 76. See also China's MDR 2024, Art. 19.

[40] Ned Feder, "Medical Devices – Balancing Regulation and Innovation" (2012) 366(3) New England Journal of Medicine 280.

[41] RM Guerra-Bretaña and AL Flórez-Rendón, "Impact of Regulations on Innovation in the Field of Medical Devices" (2018) 34(4) Research on Biomedical Engineering 356.

[42] Bruce Liu, Shiying He, and Victoria Liu, "2024 China Medtech Industry Outlook" (Simon-Kucher, 30 January 2024) <https://www.simon-kucher.com/en/insights/2024-china-medtech-industry-outlook> accessed 23 June 2025.

[43] R Tarricone and others, "An Accelerated Access Pathway for Innovative High-Risk Medical Devices under the New European Union Medical Devices and Health Technology Assessment Regulations? Analysis and Recommendations" (2023) 20(4) Expert Review of Medical Devices 259 <https://doi.org/10.1080/17434440.2023.2192868> accessed 23 June 2025.

[44] MedTech Europe, *The Future of Europe's Medical Technology Regulations* (Position Paper, November 2023) <https://www.medtecheurope.org/wp-content/uploads/2023/11/medtech-europe_future-of-medical-technology-regulations_position-paper_2023.pdf> accessed 23 June 2025.

[45] Provisions for Supervision and Administration of Medical Device Manufacturing (<医疗器械生产监督管理办法>, "Medical Device Manufacturing Provisions") (issued on 18 February 2022, effective since 1 May 2022),

devices do not require a manufacturing permit but must be filed with a city-level authority (i.e., cities with subordinate districts).[46]

China's MDR and SAMR rules also govern the "operation" (经营, jīng yíng)[47] of medical devices, a broader concept than "distribution," encompassing sales (both online and offline), procurement, warehousing, quality control, transport, and after-sales services.[48] To carry out these "operational" business activities, Class III devices require a permit. The deployment and use of medical devices is separately regulated under binding departmental rules.[49] Unlike the EU, where CE marking permits EU-wide use (though decisions concerning reimbursement for the use of a medical devices will still be made by various entities at the national level),[50] China imposes specific obligations on device deployers (e.g., hospitals) covering procurement, storage, usage, maintenance and transfer. Additionally, the clinical application of medical devices in healthcare institutions also falls under the authority of the National Health Commission (NHC).[51]

In this regard, prominent regulatory divergences between the EU and China become apparent in the post-certification stage, i.e., after a medical device is certified, the EU MDR predominantly emphasises pre-market certification and ongoing post-market surveillance, with manufacturing and distribution largely governed by integrated quality management requirements and Member State enforcement. This major distinction highlights China's more proactive but segmented regulatory approach in the post-certification stage. Unlike the EU MDR, where attaining the CE mark affords manufacturers discretion over the manner and planning of production, distribution, and sales, the Chinese approach, although designed to ensure continuous oversight, imposes a considerably greater and more complex compliance burden on manufacturers seeking access to the Chinese market after initial certification. This regulatory approach, featuring a heightened level of caution throughout the entire device lifecycle, also aligns with the Chinese government's historical tendency to exercise strong state oversight over market activities.[52] The authors of this paper would argue that these post-certification requirements represent and area where China's regulation of medical device can be

State Administration of Market Regulation <https://english.nmpa.gov.cn/2022-09/30/c_817470.htm> accessed 23 June 2025. See Art. 10.

[46] Medical Device Manufacturing Provisions 2022, Art. 22.

[47] Provisions for Supervision and Administration of Medical Device Distribution (<医疗器械经营监督管理办法>, "Medical Device Distribution Provisions") (issued on 18 February 2022 and effective since 1 May 2022), State Administration of Market Regulation <https://english.nmpa.gov.cn/2022-09/30/c_817469.htm> accessed 23 June 2025. This translation uses 'distribution' for '经营', though the authors argue "operation" is a more accurate rendering. Regarding the scope of this notion, see in particular Art. 29.

[48] There are several pieces of departmental rules and guidance-type documents governing the phase of distributing medical devices in China, including the Provisions for Supervision and Administration of Medical Device Distribution 2022 (<医疗器械经营监督管理规定>), the Administrative Measures on the Supervision of Online Sales of Medical Devices 2018 (<医疗器械网络销售监督管理办法>), the Good Practices for Medical Devices Distribution 2023 (<医疗器械经营质量管理规范>) and the Good Practices for Online Sales of Medical Devices 2023 (<医疗器械网络销售质量管理规范>).

[49] It refers to the Administrative Measures on the Supervision of Use Quality of Medical Devices 2016 (<医疗器械使用质量监督管理办法>). 99.

[50] Giuseppe Boriani and others, "Device Therapy and Hospital Reimbursement Practices Across European Countries: A Heterogeneous Scenario" (2011) 13 Suppl 2 *EP Europace* ii59.

[51] Administrative Measures for Clinical Use of Medical Devices (<医疗器械临床使用管理办法>) (issued on 12 January 2021 and effective since 1 March 2021), National Health Commission <http://www.nhc.gov.cn/wjw/c100022/202201/28815d5bc9274b3db88b70a163b1e0e1/files/f09df081829c40e2b56deaf9c52c9ecc.pdf> accessed 23 June 2025.

[52] Daolu Tang and others, "Regulatory Approaches towards AI Medical Devices: A Comparative Study of the United States, the European Union and China" (2025) 153 Health Policy 105260 https://doi.org/10.1016/j.healthpol.2025.105260 accessed 23 June 2025.

regarded as being significantly more stringent and complex than the EU in terms of both substantive and procedural requirements.

V. Software and AI-Based Medical Devices in China and the EU

1. Software As a Medical Device

China incorporated software into medical device regulation in 2002 through its catalogue of medical devices classifications. The main regulatory requirements for software as medical devices are provided in NMPA's Guiding Principles for Technical Review of Medical Devices Software Registration ("SaMD Guiding Principles"), firstly published in 2015 and revised in 2022.[53] The guidelines distinguish between software as medical device (SaMD), which runs independently on the general computing platform and accomplishes its intended medical purpose(s) without hardware, and software in medical device (SiMD), which controls medical device hardware or runs on a medical computing platform. In principle, SaMD requires stand-alone registration, while SiMD needs to be registered with its associated hardware device.

Under China's classification system for medical devices, all medical software falls under Class II or III medical devices depending on its intended use and associated risk, requiring government approval before market entry.[54] Specifically, the regulatory classification depends on the software's function and impact on medical decision-making. If used for supporting clinical decisions, such as diagnosis, treatment recommendations, or risk assessment, an AI SaMD falls under Class III,[55] requiring rigorous review and approval by the NMPA as the central regulator. In contrast, an AI SaMD designed for auxiliary functions, such as data processing and measurement to provide clinical reference information, is categorised as Class II and is subject to approval by local regulatory authorities.[56]

There are some subtle but important distinctions between the classification of software as a medical device in China and the EU. A key divergence is that software as a medical device in Europe will nearly always be classed as a highest-risk class III device,[57] whereas this need not necessarily be the case in China. This flexibility in China stems from its option of SaMD, which permits certain software-driven functions to be deemed lower risk as class II. Under the EU's MDR, however, no such option exists, and most software based medical devices are elevated to class III. According to the MDR's classification rule, class III will apply to software intended to provide information for diagnosis decision-making if "such decisions have an impact that may cause death or an irreversible deterioration of a person's state of health."[58] This criterion could be understood to encompass software that delivers data for diagnostic or treatment decisions, potentially resulting in fatal or

[53] Centre for Medical Device Evaluation of the NMPA, *Guiding Principles for Registration Review of Medical Devices Software* (revised in 2022) (<医疗器械软件注册审查指导原则>), <https://www.cmde.org.cn/flfg/zdyz/zdyzwbk/20220309091706965.html> accessed 23 June 2025.

[54] See Rules for Classification of Medical Devices (<医疗器械分类规则>) (effective since 1 January 2016) China Food and Drug Administration <https://english.nmpa.gov.cn/2019-10/11/c_415411.htm?utm_source=chatgpt.com> accessed 23 June 2025.

[55] See NMPA, Circular on the Publication of Guiding Principles for Defining the Classification of Artificial Intelligence Medical Software Products (No. 47 of 2021) (<国家药监局关于发布人工智能医用软件产品分类界定指导原则的通告（2021年第47号)>) <https://www.nmpa.gov.cn/ylqx/ylqxggtg/20210708111147171.html> accessed 23 June 2025.

[56] Ibid.

[57] European Commission Medical Device Coordination Group (MDCG), "Guidance on Qualification and Classification of Software in Regulation (EU) 2017/745 (MDR)" (MDCG 2019-11, 26 November 2019), <https://health.ec.europa.eu/system/files/2020-09/md_mdcg_2019_11_guidance_qualification_classification_software_en_0.pdf> accessed 23 June 2025.

[58] This is outlined in the rule 11) on device classification of Annex VIII of Regulation (EU) 2017/745.

permanent health damage, as well as software tracking critical physiological metrics where changes could pose an immediate risk to the patient. Consequently, software with auxiliary functions, such as data processing and measurement to provide clinical reference information, which could be classified as class II in China, is often compelled into the more strenuous class III category in the EU because a failure could, at least in theory, contribute to a decision resulting in such irreversible harm.

2. AI as a Medical Device

China's health regulators first acknowledged AI's application in healthcare in 2009, introducing two regulatory guidelines for "AI-assisted diagnosis" and "AI-assisted treatment" technologies.[59] However, these guidelines were limited in scope, addressing only auxiliary diagnostic software and clinical decision support systems while excluding AI-embedded clinical diagnostic and treatment devices. With the rapid advancement of AI technologies, the NMPA formally introduced the concept of "AI medical software" in its guiding document in 2021.[60] This term targets SaMD that leverages medical device data and artificial intelligence technology to realise its intended medical use.[61] However, not all AI medical software qualify as a medical device, especially where the software is not used for its intended medical purpose. Building on this term, the NMPA proposed "AI medical devices" in another key guideline in March 2022, which specifically addresses the registration of AI-based medical devices.[62] It explains the term "AI medical devices" in a way that aligns with the concept of "AI medical software." Key factors still lie on the utilisation of medical device data, the application of AI technology and the realisation of intended medical purpose(s). Additionally, it established standardised requirements for whole-lifecycle quality control, technical evaluations, and regulatory documentation for registering AI medical devices.

In the EU, the regulatory landscape for medical devices has recently been augmented by the introduction of the AI Act, which entered into force in 2024.[63] This horizontal legislation fills a critical gap as the MDR itself lacks any specific mentioning of the term "AI." While not exclusively targeting the health sector, the AI Act's provisions are cumulative, applying to all medical devices that have AI components or consist entirely of AI software.[64] Given the increasing integration of AI into healthcare technology, the AI

[59] Ministry of Health (replaced by the National Health Commission), Management Standards for Artificial Intelligence Assisted Diagnosis Technology (trial) (<人工智能辅助诊断技术管理规范(试行)>) (13 November 2009) <http://www.nhc.gov.cn/bgt/s10639/200911/9b687d70d88a4a8e965b7864a66ecbdb.shtml> accessed 23 June 2025; Management Standards for Artificial Intelligence Assisted Treatment Technology (trial) (<人工智能辅助治疗技术管理规范(试行)>) (13 November 2009) <http://www.nhc.gov.cn/bgt/s10639/200911/fb06b4df4fa741ad97a709674fb346bc.shtml> accessed 23 June 2025. These were lately revised by the National Health Commission in 2022.

[60] See NMPA's Guiding Principles for Defining the Classification of Artificial Intelligence Medical Software Products (No. 47 of 2021) (n 55).

[61] "Medical device data" in this guiding document refers to objective data generated by medical devices for medical purposes, such as medical image data, physiological parameter data, and in vitro diagnostic data.

[62] Center for Medical Device Evaluation of the NMPA, Circular on the Publication of Guiding Principles for Reviewing the Registration of Artificial Intelligence Medical Devices (No. 8 of 2022) (<国家药监局器审中心关于发布人工智能医疗器械注册审查指导原则的通告(2022年第8号)> <https://www.cmde.org.cn//xwdt/shpgzgg/gztg/20220309090800158.html> accessed 23 June 2025.

[63] Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 laying down harmonised rules on artificial intelligence and amending Regulations (EC) No 300/2008, (EU) No 167/2013, (EU) No 168/2013, (EU) 2018/858, (EU) 2018/1139 and (EU) 2019/2144 and Directives 2014/90/EU, (EU) 2016/797 and (EU) 2020/1828 (Artificial Intelligence Act), PE/24/2024/REV/1 [2024] OJ L 1689/1.

[64] Aboy, Mateo, Timo Minssen and Effy Vayena, "Navigating the EU AI Act: Implications for Regulated Digital Medical Products" (2024) 7(1) NPJ Digital Medicine 237 <https://doi.org/10.1038/s41746-024-01232-3> accessed 23 June 2025.

Act's role in the governance of medical devices in Europe should not be understated. The AI Act lays down key requirements for AI medical devices that apply alongside those imposed by the MDR. These include the obligation to implement a risk management system, specific rules on how a data model is to be trained (including measures to reduce bias or inaccuracy), and extensive documentation requirements.

Interestingly, China does have a few legislative initiatives for broader AI governance, though they are not as systematic as the EU's comprehensive AI Act. Where binding regulation does exist in China, it is mostly sectoral in nature and focuses on specific risks. The most prominent one is the Interim Measures on the Management of Generative Artificial Intelligence Services ("GenAI Measures"),[65] issued in 2023 by the Cyberspace Administration of China (CAC), a Chinese super cyber regulator.[66] This is often seen as a quick regulatory response to the unplanned popularisation of generative AI products at the end of 2022,[67] even preceding the full effectiveness of the EU's AI Act. Unlike the EU's risk-based approach, China's Measures concerning Generative AI primarily adopt an outcome-based model focused on content and the potential political risks from AI products.[68] The GenAI Measures oblige AI service providers to prevent from generating legally prohibited contents, such as those endangering national security, social stability, and harmful disinformation.[69] Other binding laws associated with AI address algorithmic recommendations for online content dissemination and deep synthetic technology for generating images, videos or disinformation.[70] Whilst these laws may have a bearing on some niche forms of medical device – for instance, a health and wellness chatbot using generative AI – it is likely that most will not fall under their direct application, being governed instead by the specific AI medical device guidelines issued by the NMPA.

However, the text of China's GenAI Measures does not specify its interplay with any other sector-specific regulatory frameworks, such as that for medical devices. The CAC's competence in AI governance, data protection and cybersecurity does not appear to conflict with other central regulators for healthcare, industry or technology. This suggests that the GenAI Measures has a particular motivation of content regulation rather than sector-specific compliance. As its application to AI-based medical devices is not as clear as that in the EU's AI Act (which, for example, explicitly recognises AI systems used in medical devices as high-risk systems), the authors of this paper would argue that China's GenAI Measures cannot be viewed as being equivalent to the EU's AI Act in this context. This is, in general, representative of a situation where there is no clear or strong interaction between China's medical device framework and its AI regulatory framework.

Whilst it is difficult to make generalisations in terms of a comparison between the two legal systems' treatment of AI based medical devices, the authors of this paper would make two observations. First, the EU's regulatory approach arguably entails legal certainty regarding the application of additional, potentially onerous, requirements for potential

[65] Interim Measures on the Management of Generative Artificial Intelligence Services (<生成式人工智能服务管理暂行办法>, "GenAI Measures") (issued 10 July 2023, effective since 15 August 2023), Cyberspace Administration of China <https://www.gov.cn/zhengce/zhengceku/202307/content_6891752.htm> accessed 23 June 2025.

[66] Jamie P. Horsley, "Behind the Facade of China's Cyber Super-Regulator" (*DigiChina of Stanford University*, 2022) <https://digichina.stanford.edu/work/behind-the-facade-of-chinas-cyber-super-regulator/> accessed 23 June 2025.

[67] Sara Migliorini, "China's Interim Measures on Generative AI: Origin, Content and Significance" (2024) 53 Computer Law & Security Review 105985.

[68] Gilad Abiri and Yue Huang, "A Red Flag? China's Generative AI Dilemma" (2023) 37 Harvard Journal of Law & Technology 1.

[69] GenAI Measures 2023, Art. 4.

[70] For example, the Provisions on the Administration of Algorithmic Recommendation in Internet Information Services and the Provisions on the Administration of Deep Synthesis in Internet Information Services, both issued by the CAC in 2022.

manufacturers of medical devices with AI based components. The horizontal nature of the AI Act ensures it will apply to all medical devices with an AI component, representing a clear and significant compliance burden for manufacturers in Europe. Second, and in manner that may slightly mitigate the first observation: this certainty of burden in Europe can also be viewed as a form of regulatory clarity that is currently absent in China. In the EU, manufacturers now have a clear understanding that AI-specific rules will apply to their devices in nearly all instances. This contrasts with the situation in China where it is necessary to assess the potential applicability of various fragmented and sectoral AI regulations. This distinction suggests that in theory, procedures to ensure regulatory compliance may become more homogenised and predictable in the EU than is the currently case in China.

Some scholars have described what they see as China adopting a "dual strategy that integrates stringent regulatory measures and soft guidance initiatives" for the governance of AI applications in healthcare.[71] The authors partially agree with this opinion but would like to note the dominance of sector-specific regulatory guidelines and national standards. Currently there is no specific regulatory pathway for AI-based medical devices in Chinese law, despite the NMPA and its affiliates has issued a substantive body of guidelines and national standards in this area.[72] This is because these soft regulatory guidelines establish essential requirements for the safety, effectiveness, and classification of AI-based medical devices, but do not constitute an overarching legal framework governing AI more broadly. From a practical point of view, however, these non-binding guiding documents will be important for medical device industry players to navigate an appropriate compliance strategy for AI medical devices, rather than the ambiguous and content-oriented GenAI Measures. The authors of this paper would like to point out two reasons for this. First, these sectoral guidelines and standards effectively fill the legislative gap regarding the technical assessment criteria for AI-based medical devices, especially considering their complexities compared with traditional medical devices. For example, China's industry standards have offered a clear definition of AI medical devices and formalised specific requirements for the data used in training AI medical devices,[73] which has not been included in China's existing medical device or AI regulatory framework yet. Second, these standards and guidelines also serve as the crucial basis for reviewing and approving AI-based medical devices, providing life-cycle checklists for both Chinese regulators and manufacturers. Liu and others describe it as a "rule-based" approval approach which enhances regulatory consistency, accelerates review efficiency and strengthens accountability in safety and performance assessments.[74]

VI. Non-Regulatory Factors as a Bigger Driver in China's Increased Attractiveness

In recent years, much discussion has been made of the fact that China is becoming more attractive as a place to innovate with medical devices, often at Europe's

[71] Lanyi Yu and Xiaomei Zhai, "Ethical and Regulatory Challenges of Generative Artificial Intelligence in Healthcare: A Chinese Perspective" (2024) Journal of Clinical Nursing <https://doi.org/10.1111/jocn.17493> accessed 23 June 2025.

[72] Yuehua Liu and others, "Regulatory Responses and Approval Status of Artificial Intelligence Medical Devices with a Focus on China" (2024) 7 *NPJ Digital Medicine* 255 <https://doi.org/10.1038/s41746-024-01254-x> accessed 23 June 2025. See also: Yu Han and others, "Regulatory Frameworks for AI-Enabled Medical Device Software in China: Comparative Analysis and Review of Implications for Global Manufacturer" (2024) 3 *JMIR AI* e46871 <https://doi.org/10.2196/46871> accessed 23 June 2025.

[73] Tang and others (2025) (n 52).

[74] Liu and others (2024) (n 72).

expense.[75] This discourse has frequently been accompanied by commentary bemoaning problems of "over regulation" in Europe and the brake it is placing on innovation.[76] However, while a systematic comparison of the two legal regimes is beyond the scope of this work, the analysis herein of key regulatory aspects challenges this view. This article has, by comparing some salient aspects, demonstrated that, from a purely regulatory standpoint, China does not present a clearly more advantageous environment for medical device innovators. It is therefore crucial to recognise that, although regulation exerts a significant influence, it alone does not completely explain the attractiveness of a market for medical device innovation. The authors of this paper suggest that it is consequently important to consider the legal situation as only one part of a wider context. Although it is beyond the scope of this paper to review all relevant determinants fully (and admittedly beyond the core expertise of the authors who are primarily legal scholars), it is important to acknowledge a number of other additional factors identified in various literature. These include China's market size, its dedicated R&D policies, and the level of state support. All of these are potentially significant drivers of its rising prominence in the global medical technology sector. The authors would call for further holistic empirical research in this area, taking into account these factors together with the legal issues outlined in this paper. Some of these factors are examined briefly below.

First, China offers a market of almost unparalleled scale to medical device manufacturers. Its medical device market is already substantial, accounting for approximately 25% of the Asia-Pacific market value and over 8% of the global market.[77] In the first two decades of the twenty first century, the medical device market in China expanded more than twenty-fivefold.[78] This can be contrasted with the Europe which averaged growth close to about 5 percent annually.[79] The Chinese market was furthermore projected to expand significantly, with a compound annual growth rate (CAGR) of 5.6% from 2023 to 2025. Whilst Europe's market remains larger in absolute value,[80] China's is expanding at a faster rate,[81] making it a compelling destination for medical device development and investment.

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[75] See, for example: Gerardo Fortuna, 'Commission finds evidence of discrimination against EU medtech firms in China' (*Euronews*, 14 January 2025) ≤https://www.euronews.com/health/2025/01/14/commission-finds-evidence-of-discrimination-against-eu-medtech-firms-in-china> accessed 23 June 2025; Sally Ye, "China healthcare tech insight: trends, risks and opportunities" (World Health Expo Insights, 17 November 2022) ≤https://www.worldhealthexpo.com/insights/healthcare-trends/china-healthcare-tech-insight-trends-risks-and-opportunities> accessed 23 June 2025.

[76] See, for example: Nick Paul Taylor, "EU medtech regulations reducing options for patients, companies say" (*MedTech Dive*, 13 January 2025) <https://www.medtechdive.com/news/eu-medtech-regulation-comments-mdr-ivdr/737133/> accessed 23 June 2025; Sally Burnett, "When Will the EU MDR Burden Begin to Ease for Medical Device Manufacturers?" (*MDDI Online*, 21 July 2022) <https://www.mddionline.com/regulatory-quality/europe-s-mdr-may-be-hurting-innovation> accessed 23 June 2025.

[77] Isaac Hanson, "China's Mass Surveillance May Be Key to Growth in Medical Device Sector" *Medical Device Network* (3 October 2023) <https://www.medicaldevice-network.com/news/chinas-mass-surveillance-may-be-key-to-growth-in-medical-device-sector/?cf-view> accessed 23 June 2025.

[78] Sok Teng Cheong and others, "Building an Innovation System of Medical Devices in China: Drivers, Barriers, and Strategies for Sustainability" (2020) 8 Sage Open Medicine 2050312120938218.

[79] MarketsandMarkets, "European Medical Devices Market by Type (Diagnostic Imaging, Endoscopy Equipment, Respiratory Care, Cardiac Monitoring Devices, Haemodialysis Devices, Ophthalmic Devices, Anesthesia Monitoring), End User (Hospitals, Home-care) & Region – Global Forecast to 2025" (Market report MD 7658, MarketsandMarkets, May 2020) ≤https://www.marketsandmarkets.com/Market-Reports/european-medical-devices-market-241277169.html> accessed 23 October 2025.

[80] See reports: BMI Research (Fitch Solutions), *World Medical Devices Market Factbook 2024* (summary report, 2024) <https://your.fitch.group/rs/732-CKH-767/images/worldwide-medical-devices-market-factbook-summary.pdf> accessed 23 June 2025; MedTech Europe, *European IVD Market Statistics Report 2022* (2024) <https://www.medtecheurope.org/wp-content/uploads/2022/12/european-ivd-market-report-2022.pdf> accessed 23 June 2025.

[81] See Hanson (2023) (n 77).

Second is the availability of clear state support and industrial policy incentives, exemplified by the "Made in China 2025" initiative.[82] This government strategy is designed to elevate China to a leading position in global high-tech manufacturing, with the medical device industry as a key focus. The policy aims to substantially boost the adoption of domestically manufactured medical devices in China's top hospitals, setting ambitious benchmarks to increase the share of domestically produced devices in hospitals to 50% by 2020, 70% by 2025, and 95% by 2030.[83] In Europe, by contrast, there is relatively little control of industrial policy at the EU level. Much competence remains with the Member State, leaving a fragmented policy picture.[84]

Third, the Chinese government actively promotes and funds research and development (R&D) funding. Significant state investment in the medical device sector is expanding the nation's domestic R&D capabilities, positioning China as an emerging leader in MedTech innovation.[85] Furthermore, the draft Medical Device Management Law reinforces such state support by, for example, by supporting cross-disciplinary research and development among institutions and industry, and by establishing a dedicated development fund for the medical device sector.[86] This incentivization of R&D activities not only encourages potential manufacturers to develop their products in China but also increases the likelihood that they will choose to launch their products at an early stage in the Chinese market. In Europe, whilst significant funds that are used for R&D related to medical devices are made available centrally through programmes such as Horizon, Member States retain primary responsibility for their national R&D funding programs, including grants, tax incentives and direct support for medical device companies.[87]

Fourth, China is implementing a more coordinated reimbursement policy that facilitates the uptake of innovative medical devices. The National Healthcare Security Administration, for instance, aims to have local reimbursement lists implemented across all provinces by 2025, which will aid innovators in obtaining reimbursement and enhancing product adoption.[88] This centralised effort contrasts sharply with the situation in the EU which has little direct power over reimbursement policies at member state levels. This fragmentation can make it more difficult to stimulate consistent market uptake for new technologies across the Union, even in areas deemed strategically desirable.[89]

[82] For more on the "Made in China 2025" initiative see: Alberto Gabriele, "The Made in China 2025 Plan" in *Enterprises, Industry and Innovation in the People's Republic of China* (Springer, Singapore 2020) <https://doi.org/10.1007/978-981-15-2121-8_11> accessed 23 June 2025. See also: Gemma Conroy, "How 'Made in China 2025' Helped Supercharge Scientific Development in China's Cities" Nature (7 November 2024) <https://www.nature.com/articles/d41586-024-03522-y> accessed 23 June 2025.

[83] European Commission, "COMMISSION STAFF WORKING DOCUMENT Factual findings of the IPI investigation on the procurement market for medical devices in the People's Republic of China. Accompanying the document REPORT FROM THE COMMISSION pursuant to Article 5(4) of Regulation (EU) 2022/1031 on the investigation under the International Procurement Instrument concerning measures and practices of the People's Republic of China in the public procurement market for medical devices" COM(2025) 5 final (14 January 2025) <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=celex:52025SC0002> accessed 23 June 2025.

[84] Elisa Lievevrouw, Luca Marelli and Ine Van Hoyweghen, "Weaving EU Digital Health Policy into National Healthcare Practices. The Making of a Reimbursement Standard for Digital Health Technologies in Belgium" (2024) 346 Social Science & Medicine 116620.

[85] Sok Teng Cheong et al, "Building an Innovation System of Medical Devices in China: Drivers, Barriers, and Strategies for Sustainability" (2020) 8 *SAGE Open Medicine* 2050312120938218 <https://doi.org/10.1177/2050312120938218> accessed 23 June 2025.

[86] The draft Medical Device Management Law 2024, Arts. 4 and 5

[87] Lievevrouw, Marelli and Van Hoyweghen (n 87).

[88] Bruce Liu et al., "2025 Outlook for Medtech in China" (*Simon-Kucher Insights*, 28 May 2025) <https://www.simon-kucher.com/en/insights/2025-outlook-medtech-china> accessed 23 June 2025.

[89] Richard Rak and Paul Quinn, Enhancing Digital Health Innovation in the EU with Effective Industrial Strategy Policies – A Focus on Wearable Medical Devices, CIUI, B-E (eds), Publications Office of the European Union, Luxembourg 2024, JRC138798 <https://data.europa.eu/doi/10.2760/88816> accessed 23 June 2025.

Fifth, and perhaps most controversial, is the fact that China is increasingly able to harness the power of big data on a scale that Europe cannot. Through state-led mass surveillance and big data initiatives, China is able to gather extensive health-related datasets.[90] This data fuels advancements in fields like drug discovery, clinical trial optimisation and clinical decision support, and will be crucial for training of medical devices in general and AI based medical devices in particular.[91] Whilst the EU is also developing frameworks to improve the availability of secondary health data across Member States – such as the European Health Data Space – these initiatives must operate within a much stricter legal framework of data protection and individual privacy rights, creating significant hurdles to large-scale data aggregation.

VII. Conclusion

In recent years, China's attractivity as a market for medical device innovation has been increasing relative to that of Europe, a trend projected to continue as its domestic market matures. The evolving regulatory picture is often cited as an important reason for this shift.[92] Whilst Europe's regulatory environment has become more stringent for medical devices in recent years,[93] China's framework is increasingly viewed by some as more favourable to innovation. This article seeks to probe the veracity of this perception by undertaking a focused comparison of the most salient aspects of the two regulatory frameworks. Although a systematic, point-by-point analysis is beyond the scope of this work, the authors of this article therefore would argue that the picture is far from black and white.

A number of salient factors demonstrate this mixed picture. As this article has discussed, China clearly does possess some features in its regulatory approach that may give it a competitive edge over the EU. These include, most notably, its procedure for accelerated approvals for certain medical devices that are identified as meeting a pressing clinical need. Given that time-to-market is a major concern for manufacturers, this pathway will be undoubtedly appealing. Other important aspects include the fact that it may be possible to obtain a lower classification (i.e., below Class III) for software and AI-based medical devices more often in China than in the EU (where the vast majority of devices with software will be classified as Class III). Furthermore, China does not have a generalised AI regulatory farmwork equivalent to the EU's AI Act that will apply to all forms of medical devices that incorporate AI. This absence of a cumulative regulatory layer means that the compliance burden on many AI-enabled medical devices will arguably be lower in China.

Conversely, the structure of the Chinese regulatory system presents several potential disadvantages when compared to the EU system, mote notably concerning legal certainty. In the EU, for instance, the legal framework is established by a "Regulation," a legislative instrument that occupies a clear and unrivalled position in legislative hierarchy, subordinate only to the EU treaties. This contrasts sharply with the situation in China, where the main instrument governing medical devices does not take the form of national

[90] Jun Wu et al., "Application of Big Data Technology for COVID-19 Prevention and Control in China: Lessons and Recommendations" (2020) 22 Journal of Medical Internet Research e21980 <https://doi.org/10.2196/21980> accessed 23 June 2025.

[91] Zhihui Zhang and Wei Rao, "Key Risks and Development Strategies for China's High-End Medical Equipment Innovations" (2021) Risk Management and Healthcare Policy 3037 <https://doi.org/10.2147/RMHP.S306907> accessed 23 June 2025.

[92] Xu Song et al., "Advancing Medical Device Regulatory Reforms for Innovation, Translation and Industry Development in China" (2022) 37 Journal of Orthopaedic Translation 89, doi: 10.1016/j.jot.2022.09.015.

[93] Amanda Pedersen, "Europe's MDR May Be Hurting Innovation" (MD+DI, 21 July 2022) ≤https://www.mddionline.com/regulatory-quality/europe-s-mdr-may-be-hurting-innovation> accessed 24 June 2025.

laws (i.e., created by the highest legislature) but rather that of an administrative regulation. As this latter category enjoys a lower hierarchical value, it creates a degree of legal uncertainty for manufacturers; in the case of a clash with a national law of a higher rank, the administrative regulation would have to give way.

Another key factor of the Chinese system lies in its approval process for medical devices, particularly, the bodies which approval must be sought from. Whereas Europe's framework is decentralised and relies on a range of non-state actors (i.e., in the form of "notified bodies"), in China this function is carried out by more centralised, state-based agencies. While a centralised state-run system might imply simplicity, the Chinese model arguably presents several disadvantages vis-à-vis that of Europe. The EU's highly heterogeneous system of notified bodies, for instance, arguably allows for more opportunities for specialisation and competition than is the case with a more centralised system. Notified bodies can tailor their approvals processes to certain types of devices, offering expertise and choice to manufacturers. With a centralised system such diversification is not possible. Furthermore, a centralised model does not offer alternative avenues for review; in Europe, if a notified body is overloaded, manufacturers can approach another with such a vast array of reviewing structures. This distribution of capacity should, in theory, result in more efficient review timelines than a single, potentially overburdened state system. Moreover, the benefits that may come with centralisation in China are partly eroded by the existence of a range of state bodies tasked with regional and sectoral bodies, which are charged with carrying out different approval tasks depending on the classification of medical devices. For potential manufacturers unfamiliar with the particularities of the Chinese system, such complexity is likely to be unattractive. This situation is further complicated by the necessity to obtain separate types of licences for manufacturing, distribution and deployment – a lifecycle regulatory approach that contrasts sharply with the single approval sufficient for market access in the EU.

Above all, the authors of this paper would argue that differences in regulatory approaches themselves do not explain the perceived shift in attractiveness towards China for medical device innovation. Whilst the Chinese regulatory system presents certain regulatory advantages, such as pathways for accelerated approval, it is in other respects arguably more complex and less transparent than its European counterpart. In understanding why China's ecosystem as a whole is becoming more attractive, it is therefore necessary not only to consider a host of other factors but to consider them holistically. These include the increasing importance of China as a giant market for medical devices, the availability of significant state support for R&D, powerful incentives to "onshore" the production of medical devices, and the coordinated ability to stimulate national reimbursement policies that will boost the uptake of certain types of medical devices. Further consolidated study into the interplay of these factors would not only allow for a nuanced understanding of how China may be boosting its attractiveness, but perhaps also provide valuable lessons for how Europe might enhance its own competitiveness in the global MedTech landscape.

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